



National University of Science and Technology:

Department: Electrical Engineering

School: SEECS

Section: BEE-17-B:

Lab demonstrator: Eng Rafia Ahmed

Semester project: Proximity Sensor

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Introduction:

This project is based on the design and construction of a proximity sensor using the LM358 operational amplifier IC. The main purpose of this project is to detect nearby objects without touching them. The circuit works using infrared light, which reflects from an object and is then detected by the sensor. When an object comes close to the sensor, the circuit provides a clear indication using an LED and a buzzer. This project includes circuit design, simulation, practical implementation on a breadboard, and analysis of the results are obtained.

Design and implementation:

➤ Hardware Components Used

The following components were used to build the circuit:

- LM358 IC
- IR LED (transmitter)
- Photodiode / IR receiver
- Potentiometer
- Resistors
- LEDs
- Buzzer
- Breadboard
- Connecting wires
- DC power supply (9V)

Each component was chosen to ensure proper operation and safe working of the circuit.

➤ Circuit Diagram and Schematic:

The circuit was first designed and tested using Proteus software. After successful simulation, the same circuit was implemented on a breadboard. The LM358 IC is used as a comparator to compare the sensor voltage with a reference voltage set by the potentiometer.

➤ Inputs and Outputs:

Inputs:

- DC supply voltage
- Reflected infrared light from nearby objects

Outputs:

- LED turns ON when an object is detected
- Buzzer produces sound as a warning signal

➤ Implementation Procedure:

- The LM358 IC was placed properly on the breadboard.
- The IR LED and photodiode were connected at the input side.
- A potentiometer was connected to adjust the sensitivity of the circuit.
- LEDs and buzzer were connected to the output.
- Power supply was applied to the circuit.
- Objects were moved near the sensor to check the response.

Results and Analysis:

➤ Simulation Results:

During simulation, it was observed that when an object came near the sensor, the output of the LM358 changed and activated the LED and buzzer. The circuit behaved as expected in the simulation environment.

➤ Experimental Results

In practical testing, the circuit successfully detected nearby objects. The LED and buzzer responded quickly. The detection range could be adjusted using the potentiometer.

➤ Analysis and Discussion

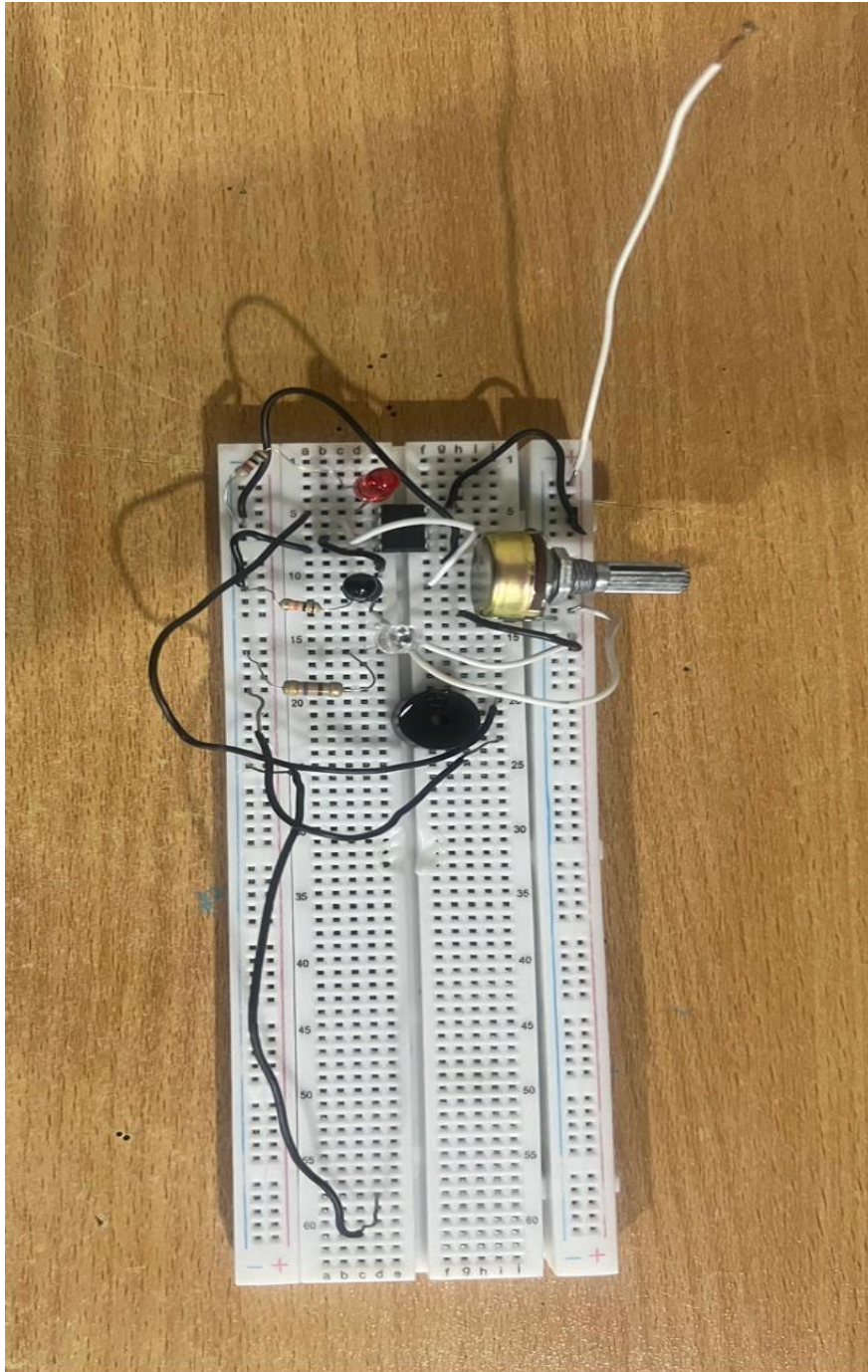
Small differences were noticed between the simulation and practical results. These differences were mainly due to ambient light, slight misalignment of the IR components, and normal variations in electronic components. However, the overall performance of the circuit was satisfactory.

Appendices:

➤ Components values:

- Resistors:
(1k,10k,470ohm)
- Potentiometer(10k)

➤ Circuit diagram on bread board:



Simulation:

